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M3-R5.1
Programming & Problem Solving Through Python
Syllabus

1. Mark Distribution

Module Unit	Written Marks (Max.)
1. Introduction to Programming, Algorithm and Flowcharts to solve problems	20
2. Introduction to Python, Operators, Expressions and Python Statements, Sequence data types	30
3. Functions, File Processing, Modules	40
4. NumPy Basics	10
5. Total	100

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ओ लेवल के तीसरे पेपर (M3-R5 /Python) मे कुल 4 यूनिट है जिन्हे 9 चैप्टर मे विभाजित किया गया है -

1.) Introduction to Programming

- The basic Model of computation
- Algorithms &Flowcharts
- Programming Languages
- Compilation
- Testing & debugging
- Documentation

2.) Algorithms and Flowcharts to Solve Problems

- Flow Chart Symbols
- Basic algorithms/flowcharts for sequential processing, decision-based processing, and iterative processing
- Examples: Exchanging values of two variables
- Examples: Summation of a set of numbers
- Examples: Decimal Base to Binary Base conversion
- Examples: Reversing digits of an integer

- Examples: GCD (Greatest Common Divisor) of two numbers
- Examples: Test whether a number is prime
- Examples: Factorial computation
- Examples: Fibonacci sequence
- Examples: Evaluate ‘sin x’ as sum of a series
- Examples: Reverse order of elements of an array
- Examples: Find the largest number in an array
- Examples: Print elements of an upper triangular matrix

3.) Introduction to Python

- Python Introduction
- Technical Strength of Python
- Introduction to Python Interpreter and program execution
- Using Comments
- Literals
- Constants
- Python’s Built-in Data types
- Numbers (Integers, Floats, Complex Numbers, Real, Sets)

- Strings (Slicing, Indexing, Concatenation, other operations on Strings)
- Accepting input from Console
- Printing statements
- Simple 'Python' programs

4.) Operators, Expressions and Python Statements

- Operators
- Types of operators
- Conditional statements: if, if-else, if-elif-else; simple programs
- Notion of iterative computation and control flow
- range function
- While Statement
- For loop
- break statement
- Continue Statement
- Pass statement
- else and assert statements

5.) Sequence Data Types

- Lists, tuples, and dictionary
- Slicing, Indexing, Concatenation, and other operations on Sequence datatype
- Concept of mutability
- Examples: finding the maximum, minimum, mean
- Examples: linear search on list/tuple of numbers
- Examples: counting the frequency of elements in a list using a dictionary

6.) Functions

- Top-down approach of problem solving
- Modular programming and functions
- Function parameters
- Local & Global variables
- The Return statement
- DocStrings
- Global statement

- Default argument values
- Keyword arguments
- VarArgs parameters
- Library functions: input(), eval(), print()
- String Functions: count(), find(), rfind(), capitalize(), title(), lower(), upper(), swapcase(), islower(), isupper(), istitle(), replace(), strip(), lstrip(), rstrip(), split(), partition(), join(), isspace(), isalpha(), isdigit(), isalnum(), startswith(), endswith(), encode(), decode()
- String operations: Slicing, Membership, Pattern Matching
- Numeric Functions: eval(), max(), min(), pow(), round(), int(), random(), ceil(), floor(), sqrt()
- Date & Time Functions
- Recursion

7.) File Processing

- Concept of Files
- File opening in various modes and closing of a file
- Reading from a file

- Writing onto a file
- File functions: open(), close(), read(), readline(), readlines(), write(), writelines(), tell(), seek()
- Command Line arguments

8.) Scope and Modules

- Scope of objects and Names
- LEGB Rule
- Module Basics
- Module Files as Namespaces
- Import Model
- Reloading Modules

9.) NumPy Basics

- Introduction to NumPy
- ndarray
- datatypes
- array attributes
- array creation routines

- Array From Existing Data
- Array From Numerical Ranges
- Indexing & Slicing

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